Arnav Mahadev

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Education

University of California, Berkeley — Berkeley, CA
Bachelor of Science in Electrical Engineering and Computer Sciences - ***GPA: 3.7/4.0*** — *Expected May 2028*

Relevant Coursework

Machine Learning, Data Structures, Efficient Algorithms, Operating Systems, Database Systems, Computer Architecture, Internet Architecture, Discrete Math, Probability Theory, Linear Algebra, Multivariable Calculus, Circuits & Devices

Experience

LG NOVA — Aug 2026 – Present
Software Engineering Intern — *Santa Clara, CA*

- Root-caused silent data loss in a platform-wide Flask/PostgreSQL endpoint capping all paginated downloads at **500 rows** and shipped a keyset pagination redesign guaranteeing complete, deterministic results
- Deployed fix across Python and React/TypeScript repos, expanding semantic search coverage of PitchBook financial fields by **233%**, backed by a **38-test suite** verifying gap-free pagination across **1,200+ rows**

iPick.ai — Aug 2026 – Present
Machine Learning Engineering Intern — *Berkeley, CA*

- Built an LLM-based portfolio optimizer using **Claude API** that segments holdings into investment tracks, prunes dominated positions, and provides per-stock and per-portfolio risk metrics
- Engineered a **reinforcement learning pipeline** of three trading agents (power picks, hidden gems, and a short-target detector) that uses realized returns over a fixed holding window as the reward signal to update each agent's policy

Software Engineering Intern — Jan 2026 – May 2026

- Architected a graph-based company relationship visualizer using **React, D3.js, and PostgreSQL**
- Extracted supplier and subsidiary relationships across **5,000+ public company tickers**, storing their structured edges in **PostgreSQL** to power graph-based visualization in the frontend
- Integrated the **Claude API** to generate real-time news summaries, enriching company profiles with live information

Maadhev LLC — May 2026 – Aug 2026
Software Engineering Intern — *Cupertino, CA*

- Built a custom ACH payment system (Python, Flask, Stripe API) integrated with a Shopify app, enabling bank transfer payments at **70% lower per-transaction cost** than card processing
- Automated order confirmation via **Stripe webhook handlers** that sync payment and fulfillment data between Stripe and Shopify in real time, eliminating manual reconciliation across every order

Projects

Falsetto | *Python, PyTorch, Hugging Face Transformers, scikit-learn, Gradio, Modal, librosa* — Jul 2026 – Sep 2026

- Developed an AI-generated music detector using MERT embeddings and an attention-based AudioCAT classifier
- Achieved **87.5% accuracy and 0.972 AUC** with an end-to-end ML pipeline, validated by **56 automated tests**

SoccerBoard | *Python, JavaScript, XGBoost, PyTorch, FastAPI, Docker* — May 2026 – Jul 2026

- Trained a match predictor backtested on **2,500+ matches**, improving outcome accuracy from **47.5% to 59.9%**
- Developed an expected goals model matching StatsBomb's professional benchmark **within 3.2%** of actual goals

SaveBites | *JavaScript, Python, React, HTML/CSS* — Sep 2025 – Dec 2025

- Built the grocery-tracking backend for a food-waste app that scans receipts to auto-populate ingredient inventory
- Developed a **RESTful API** for full CRUD on **MongoDB** ingredient data, fed by an **OCR pipeline** for receipts

Leadership

Vice President of Professional Development | *Open Project Berkeley* — Aug 2025 – Present

- Ran resume workshops, mock interviews, and **50+ 1:1 resume reviews** for a club of **300 technical consultants**

Soccer Captain | *De Anza Force 06B NPL & Monta Vista Varsity Soccer* — Aug 2021 – Jun 2024

- Captained my school and club teams, scored **40+ goals** over three seasons as **3x All-League and league top scorer**

Eagle Scout | *Scouts BSA Troop 476* — Mar 2017 – Feb 2024

- Led a classroom painting project, managing **15 volunteers**, **mentored 5 scouts** and completed **200 service hours**

Technical Skills

Languages: Python, JavaScript, Java, C#, SQL
Backend: Node.js, Express, Flask, FastAPI, REST
Frontend: React, HTML/CSS, Liquid
ML/Data: XGBoost, PyTorch, scikit-learn, NumPy, pandas
Data/Cloud: PostgreSQL, MongoDB, AWS, Docker
Tools: Git, pytest, JUnit